Volatility Forecasting with Machine Learning: A Horse Race Across GARCH, HAR, and Tree-Based Models

Akram Khan*

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Abstract

This paper runs a controlled horse race across seven volatility-forecasting configurations on S&P 500 realised volatility from January 2004 through November 2025. The seven are three GARCH variants (GARCH, EGARCH, GJR-GARCH), the HAR-RV model of Corsi (2009), two gradient-boosted trees (LightGBM, XGBoost), and an equal-weight ensemble of LightGBM, HAR-RV, and GARCH(1,1). Every model is evaluated on the same data, the same features, and the same expanding-window protocol over a January 2022–November 2025 test period of 980 trading days. The test window spans the 2022 bear market, the 2023 rally, the 2024–2025 rate-normalisation regime, and a 19.4% drawdown year. QLIKE serves as the primary loss function, alongside RMSE and Mincer-Zarnowitz R^2. Diebold-Mariano two-sided p-values are reported for all key pairs, plus the Hansen-Lunde-Nason Model Confidence Set, the Hansen Superior Predictive Ability test, and the White Reality Check, all computed with the open-source `vol-eval` package of Khan (2026b).

Three findings emerge. First, on the full test sample no model separates from the field at conventional significance levels. The equal-weight ensemble has the lowest point-estimate QLIKE (0.3431); GJR-GARCH is the best individual model (QLIKE 0.3447); the Diebold-Mariano test cannot reject equal expected loss between them ($p = 0.90$). The ensemble's narrow lead is a precision result, not a power result. Second, the full-sample ranking masks a substantial subperiod reversal. In 2022 the GARCH family wins, with EGARCH best at QLIKE 0.2346 and LightGBM last. In 2023–2025 the tree models take the lead and HAR-RV drops to last. The 2022 environment is structurally out-of-distribution for models

*Independent Researcher. ORCID: 0009-0002-7521-8648. Email: 1819ak@gmail.com. SSRN: ssrn.com/abstract=6663418. Replication code, data, and results: github.com/ayk5511/volatility-forecasting.

trained on 2004–2021, and the parametric GARCH variants adapt more gracefully than the 30-feature trees. Third, the LightGBM split-importance ranking shows VIX as the dominant feature; the leverage effect is captured through three different return-based features that together account for roughly 19% of split count; and long-horizon RV signals dominate the short-horizon ones. The full LaTeX source, the data-collection and modelling scripts, the per-day forecasts of all seven models, and the supplementary metrics are released under permissive licences, scoring 2 on the Reproducibility Disclosure Score rubric of Khan (2026a).

Keywords: volatility forecasting, realized volatility, GARCH, HAR-RV, machine learning, LightGBM, XGBoost, forecast combination, S&P 500

JEL Classification: C22, C45, C53, G17

Contents

1 Introduction

Volatility forecasting is one of the few areas in finance where the question "does this model actually work?" has a fairly clean answer. Return prediction is the opposite case: the signal-to-noise ratio is so low that statistical significance is routinely debated. Volatility, by contrast, is persistent, clustered, and measurable. A model that forecasts tomorrow's volatility well is immediately useful for option pricing, risk management, portfolio construction, and margin setting. The practical stakes are high, and the evaluation criteria are well-established.

What is less settled is which model works best. The field has accumulated a large zoo of approaches over the past three decades. At one end sit parametric time-series models: the GARCH family of Bollerslev (1986) and its many extensions (EGARCH, GJR-GARCH, FIGARCH, Realized GARCH, and others). In the middle sits the Heterogeneous Autoregressive model for realized volatility (HAR-RV) of Corsi (2009), which achieves surprisingly good forecasting performance with nothing more than three lagged moving averages of realized volatility. At the other end sit machine-learning methods: gradient-boosted trees, recurrent neural networks, and, more recently, transformers and attention-based architectures.

Each new paper in the space typically compares its proposed method against one or two baselines and reports a win. The result is a literature that is hard to synthesise. Does LightGBM beat GARCH? It depends on the market, the sample period, the horizon, the loss function, and how carefully the author tuned the GARCH specification. Cross-study comparisons are close to meaningless because no two papers use the same data, features, or evaluation protocol.

This paper runs the horse race that the literature needs. We implement six models. Three GARCH variants (GARCH, EGARCH, GJR-GARCH); HAR-RV; LightGBM; XGBoost. We add a simple equal-weight ensemble. All seven are evaluated on the same data, the same features, the same train/test split, and the same loss functions.[1] The data is the S&P 500, which we chose not because US equities are the most interesting market but because they are the most studied. That makes our results directly comparable to prior work. The sample runs from January 2004 through November 2025, which gives us 5,446 usable trading days after feature construction. The test period starts in January 2022, so every model has to forecast through the 2022 bear market, the 2023 AI-driven rally, and the 2024–2025 rate-normalisation regime. That is a deliberately challenging out-of-sample environment for models trained on pre-pandemic data.

[1] We initially planned to include LSTM and GRU recurrent networks. The PyTorch backend in our development environment had compatibility issues with the Python 3.14 toolchain we used elsewhere in the pipeline, and rather than introduce a separate environment for one model class we chose to defer the neural-network comparison. We flag this as a limitation in Section 6; the question of whether properly implemented sequence models add value over the methods we test here remains open.

Why we wrote this paper

Three concerns motivate the study, each addressing a habit in the existing literature that we think is worth changing.

Single-number reporting can mislead. Most ML-vs-classical comparisons in this subfield report a full-sample QLIKE (or RMSE, or R^2) and stop. A reader has no way to tell whether the reported winner survives different regimes, holds up under statistical testing, or merely averages over heterogeneous environments that pull in different directions. We will show that the full-sample ranking in our own data hides a model-class reversal between 2022 and 2023–2025, and that several ostensible "wins" in the full-sample table do not survive a Diebold-Mariano test. We argue that any forecasting comparison in this space should report at minimum a full-sample table, a subperiod or regime-conditional table, and pairwise statistical significance.

Regime robustness, not average accuracy, is what risk managers buy. The applied audience for these models is not a researcher checking a leaderboard; it is a risk manager who has to set initial margin tomorrow and a model-validation team that has to defend the choice next quarter. A model that wins by a wide margin in calm markets but fails in stressed ones is worse, in their use case, than a model that is never best but never disastrous either. Our test period (2022–2025) is constructed precisely to discriminate models on this property, and our conclusion that a simple equal-weight ensemble is the regime-robust choice is the result that matters most for that audience.

Reproducibility in this subfield is poor and we want to push back on that. Cross-study comparisons in volatility-ML are nearly impossible because no two papers share data, features, or evaluation protocol. We release the entire pipeline (raw Yahoo Finance pull through final metric tables) under permissive licenses, scoring 2 on the Reproducibility Disclosure Score rubric proposed in Khan (2026a). Anyone can rerun the comparison with a different feature set, an additional model, or a longer test period, and we explicitly invite such extensions through the repository's issue tracker.

Concrete contributions

1. A head-to-head comparison of seven model configurations (three GARCH variants, HAR-RV, LightGBM, XGBoost, and an equal-weight ensemble) on post-pandemic US equity volatility data, evaluated on the same data, features, train/test split, and protocol.

2. Three loss-function reports per model: QLIKE (Patton, 2011), RMSE, and Mincer-Zarnowitz R^2, accompanied by Diebold-Mariano two-sided p-values for all key model pairs with HAC (Newey-West) standard errors. Many ML papers in this space report only RMSE; we show that QLIKE and the DM test materially change the conclusions

one can defend.

3. A subperiod analysis along two cuts: calendar year (2022 versus 2023–2025) and realized-volatility regime (top quartile versus the lower three). The subperiod tables document a model-class reversal that the full-sample ranking hides.

4. A demonstration that the equal-weight ensemble (LightGBM + HAR-RV + GARCH) is statistically indistinguishable from the best individual model on the full sample (DM $p = 0.90$ vs. GJR-GARCH) yet is more robust across regimes. We argue this is the practically useful version of the *ensembles win* result.

5. Full reproducibility: code, raw inputs, the engineered feature panel, the per-day forecasts of all seven model configurations, and the metric files used to populate every table in this paper are released at `github.com/ayk5511/volatility-forecasting` under permissive licenses.

The rest of the paper is organized as follows. Section 2 reviews the volatility forecasting literature with a focus on the ML methods introduced since 2015. Section 3 describes our data and the construction of realized volatility measures. Section 4 details each forecasting model and our evaluation protocol. Section 5 presents the main results. Section 6 reports the subperiod analysis and Diebold-Mariano significance tests, and discusses limitations. Section 7 concludes.

2 Related Literature

Volatility forecasting has a deep literature, and we do not attempt to survey it here; Poon and Granger (2003) and Andersen et al. (2006) provide book-length treatments, and Khan (2026a) surveys the post-2015 ML applications more broadly. Instead, we focus on the developments most directly relevant to our comparison: the GARCH family, realized volatility models, and the ML approaches that have emerged since roughly 2015.

2.1 GARCH Models

The GARCH(1,1) model of Bollerslev (1986) remains the workhorse of conditional volatility modeling. Its asymmetric variants, the EGARCH of Nelson (1991) and the GJR-GARCH of Glosten et al. (1993), capture the leverage effect (negative returns increase volatility more than positive returns of the same magnitude), a well-documented empirical regularity. Hansen and Lunde (2005) conduct a large-scale comparison of 330 GARCH variants and conclude that no model consistently dominates the simple GARCH(1,1), a finding that influenced our decision to include the standard specification alongside EGARCH and GJR-GARCH rather than searching for exotic variants.

A limitation of daily GARCH models is that they use only closing prices, discarding the information in intraday price movements. This motivates the realized volatility approach.

2.2 Realized Volatility and the HAR Model

Realized volatility (RV), computed as the sum of squared intraday returns, provides a more accurate measure of true volatility than GARCH-implied estimates (Andersen et al., 2003). The Heterogeneous Autoregressive (HAR) model of Corsi (2009) forecasts RV using three components (daily, weekly, and monthly lagged RV), motivated by the heterogeneous market hypothesis that traders at different horizons contribute to different volatility dynamics. Despite its simplicity (it is just a linear regression with three regressors), the HAR model is remarkably hard to beat. Bollerslev et al. (2016) extend HAR with jump components and find modest improvements. Patton and Sheppard (2015) show that the HAR structure captures the long-memory properties of volatility well.

When intraday data is unavailable (as is the case for many practitioners and for longer historical samples), daily proxies for realized volatility can be constructed from daily OHLC prices using range-based estimators (Parkinson, 1980; Garman and Klass, 1980). We use both the standard close-to-close squared return and the Parkinson range estimator in our feature set.

2.3 Machine Learning Approaches

The application of ML to volatility forecasting has grown rapidly since 2015, though the literature is more scattered than for return prediction.

2.3.1 Neural Networks

Bucci (2020) compares LSTM networks to GARCH and HAR-RV for S&P 500 realized volatility and finds that LSTMs improve on GARCH during high-volatility episodes but show marginal gains in calm markets. Kim and Won (2019) apply attention-augmented LSTMs to volatility forecasting on Korean equity data. Rahimikia and Poon (2021) show that combining macroeconomic text data with neural network models improves volatility forecasts, though disentangling the contribution of the model from the contribution of the data is difficult.

A consistent theme in this literature is that neural networks struggle to beat simple linear baselines (specifically HAR-RV) when evaluated on the same data and features. The gains, when they exist, tend to concentrate in high-volatility regimes where nonlinearity matters most.

2.3.2 Tree-Based Methods

Gradient-boosted trees have received less attention for volatility forecasting than for return prediction, but the existing evidence is positive. Mittnik et al. (2022) benchmark LightGBM against 14 other methods on 20 equity indices and find that LightGBM achieves the best average performance, with the advantage concentrated in forecasting regime transitions. Luong and Dokuchaev (2021) report similar findings using random forests with a large feature set that includes options-implied features.

The appeal of tree-based methods for volatility forecasting is that they naturally capture interactions between features (e.g., the interaction between lagged volatility and VIX level may be informative) without requiring explicit feature engineering.

2.3.3 Forecast Combination

A finding that holds across the forecasting literature, not just in finance, is that combining forecasts from multiple models typically outperforms any individual model (Timmermann, 2006). For volatility specifically, Patton (2019) show that simple equal-weight combinations of volatility forecasts are competitive with sophisticated combination schemes based on past performance. This motivates our inclusion of a simple ensemble as a benchmark.

2.4 Gap in the Literature

To our knowledge, no existing study compares GARCH, HAR, and tree-based model families head-to-head on the same post-pandemic data using QLIKE as the primary evaluation metric. This is the gap we fill.

3 Data

3.1 Sample

We use daily data for the S&P 500 index (ticker `^GSPC`) and the CBOE Volatility Index (VIX, ticker `^VIX`) from January 2, 2004 through November 26, 2025, obtained from Yahoo Finance via the included `01_collect_data.py` script. The sample contains 5,534 trading days before feature construction and 5,446 usable observations after accounting for the rolling windows needed to compute lagged features and forward targets.

We divide the sample into three periods:

Table 1: Sample periods

Period	**Dates**	**Trading days**	**Purpose**
Training	2004–2018	∼3,770	Model fitting
Validation	2019–2021	∼756	Hyperparameter tuning, early stopping
Test	2022–2025	980	Out-of-sample evaluation

The test period is deliberately set to begin in January 2022. This choice is motivated by two considerations. First, it ensures that all models are evaluated on data that is genuinely out of sample: no model sees any post-2021 data during training or validation. Second, the 2022–2025 period is a challenging forecasting environment, including the 2022 bear market (S&P 500 down 19.4%), the 2023 recovery, the 2024 rate-cut expectations rally, and the 2025 post-election regime. A model that performs well in this period has demonstrated robustness to regime change.

3.2 Volatility Measures

We construct realized volatility from daily returns following standard practice:

Log returns. Daily log returns are computed as $r_t = \ln(P_t/P_{t-1})$, where P_t is the closing price on day t.

Realized volatility. For a given window of h days, realized volatility is:

$$\mathrm{RV}_t^{(h)} = \sqrt{\frac{252}{h} \sum_{i=0}^{h-1} r_{t-i}^2} \tag{1}$$

where the $\sqrt{252}$ factor annualizes the measure. We compute RV at three horizons: $h = 5$ (weekly), $h = 22$ (monthly), and $h = 66$ (quarterly).

Parkinson range estimator. As an alternative daily volatility proxy that uses more of the available price information:

$$\sigma_t^{\mathrm{Parkinson}} = \sqrt{\frac{1}{4 \ln 2} \left(\ln \frac{H_t}{L_t} \right)^2} \tag{2}$$

where H_t and L_t are the daily high and low prices (Parkinson, 1980).

Forecast target. Our primary forecast target is the 5-day forward realized volatility, $\mathrm{RV}_{t+1:t+5}^{(5)}$. We also report results for the 1-day and 22-day horizons in the robustness section.

3.3 Summary Statistics

Table 2: Summary statistics (full sample, 2004–2025)

	Mean	**Std**	**Min**	**Q25**	**Median**	**Max**
Daily log return	0.0003	0.0120	−0.1277	−0.0041	0.0007	0.1096
RV (5-day, ann.)	0.1481	0.1190	0.0097	0.0785	0.1184	1.4268
RV (22-day, ann.)	0.1562	0.1081	0.0372	0.0946	0.1261	0.9355
VIX (close)	19.02	8.55	9.14	13.45	16.59	82.69

Several features of this data are worth noting. First, the distribution of realized volatility is heavily right-skewed: the mean (14.8%) is well above the median (11.8%), reflecting occasional volatility spikes (the GFC in 2008, the COVID crash in March 2020, and the 2022 sell-off). Second, the VIX, which represents the market's expectation of 30-day forward volatility implied by S&P 500 options, is on average higher than 22-day realized volatility, consistent with the well-documented variance risk premium. Third, the maximum 5-day RV of 142.7% (annualized) occurred during March 2020, which we deliberately include in the training set so that models are exposed to at least one extreme stress event during fitting.

3.4 Feature Construction

We construct 30 features for the ML models, all strictly backward-looking to avoid look-ahead bias. The feature panel decomposes as follows (counts in parentheses sum to 30):

- **HAR components (3):** Daily ($\mathrm{RV}_{t-1}^{(1)}$), weekly ($\mathrm{RV}_{t-5:t-1}^{(5)}$), and monthly ($\mathrm{RV}_{t-22:t-1}^{(22)}$) lagged realized volatility.
- **Lagged RV (6):** Lags 1, 2, 3, 5, 10, and 22 of 5-day realized volatility.
- **Lagged absolute returns (6):** Lags 1, 2, 3, 5, 10, and 22 of daily absolute returns.
- **VIX features (4):** Same-day VIX close, 1-day-lagged VIX, 5-day moving-average VIX, and VIX daily change.
- **Asymmetry (2):** Cumulative negative and positive returns over the past 5 days, capturing the leverage effect.
- **Range volatility (1):** The Parkinson high–low estimator.
- **Volume (2):** Volume daily change and volume relative to its 22-day moving average.

- **Calendar (5):** Day-of-week indicators (Monday through Friday); included as separate dummies for the LightGBM/XGBoost split logic rather than dropping a reference level.
- **Daily log return (1):** The day-t log return r_t enters as a level feature alongside the lagged absolute-return panel.

The GARCH models use only the return series (they estimate their own conditional variance), while the HAR model uses only its three canonical components. The tree-based models (LightGBM, XGBoost) use the full feature set.

4 Methodology

We describe each forecasting model and then detail our evaluation protocol.

4.1 GARCH Family

We estimate three daily GARCH specifications using the `arch` Python package (Sheppard, 2023). All three model the conditional variance σ_t^2 as a function of lagged squared returns and lagged variance.

GARCH(1,1):

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 \tag{3}$$

EGARCH(1,1) (Nelson, 1991):

$$\ln \sigma_t^2 = \omega + \alpha \left(\frac{|r_{t-1}|}{\sigma_{t-1}} - \sqrt{2/\pi} \right) + \gamma \frac{r_{t-1}}{\sigma_{t-1}} + \beta \ln \sigma_{t-1}^2 \tag{4}$$

GJR-GARCH(1,1) (Glosten et al., 1993):

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}_{r_{t-1}<0}) r_{t-1}^2 + \beta \sigma_{t-1}^2 \tag{5}$$

where γ captures the leverage effect: negative returns ($\mathbf{1}_{r_{t-1}<0} = 1$) receive a larger weight.

GARCH models are refit every 44 trading days (approximately bimonthly) on an expanding window of returns. The conditional-variance filter at time t uses only returns observed through $t-1$, so there is no look-ahead in the filtering step. The fitted parameters, however, are estimated by quasi-maximum likelihood on the window through the end of each refit batch, which means the parameters do incorporate batch-period information. A strict walkforward design (parameters refit at the start of each batch and held fixed through it) is more conservative; we adopt the within-batch fit because parameter drift over a 44-day window is small and the simplification is common in the GARCH-evaluation literature. One-step-ahead variance forecasts are converted to annualized volatility via $\hat{\sigma}_t^{\text{ann}} = \hat{\sigma}_t \sqrt{252}$.

Refit-cadence asymmetry. The GARCH cadence (44 days) is shorter than the HAR and tree cadence (66 days). The asymmetry is deliberate. GARCH parameters $(\omega, \alpha, \beta, \gamma)$ govern conditional-variance *persistence*; a stale parameter set during a regime shift is a known weakness of the family. Refitting bimonthly limits the window during which the parameter set can drift out of step with a high-vol episode and stays standard for the comparison literature. The HAR coefficients (three) and the tree feature panels (30 features) are estimated on enough observations that quarterly refitting balances responsiveness against the cost of retraining. The asymmetry could in principle alter rankings; the natural sensitivity check is to rerun the GARCH family at a 66-day cadence, which we treat as follow-up work rather than a rerun that affects the headline numbers reported here.

4.2 HAR-RV

The Heterogeneous Autoregressive model (Corsi, 2009) is a linear regression:

$$\mathrm{RV}_{t+1:t+h}^{(h)} = c + \beta_d \, \mathrm{RV}_t^{(1)} + \beta_w \, \mathrm{RV}_{t-4:t}^{(5)} + \beta_m \, \mathrm{RV}_{t-21:t}^{(22)} + \varepsilon_t \tag{6}$$

We estimate this by OLS with an expanding window, refitting every 66 trading days (quarterly). The beauty of HAR is that it captures the long-memory properties of realized volatility through a parsimonious three-parameter structure, without requiring fractional integration.

4.3 LightGBM

LightGBM (Ke et al., 2017) is a gradient-boosted decision tree algorithm that splits data leaf-wise rather than level-wise, achieving faster training with comparable accuracy. We use the full 30-feature set described in Section 3. Hyperparameters are:

Table 3: LightGBM hyperparameters

Parameter	**Value**
Number of leaves	31
Learning rate	0.05
Feature fraction	0.8
Bagging fraction	0.8
Max boosting rounds	500
Early stopping rounds	30

Early stopping is determined using the validation set (2019–2021). The model is refit every 66 trading days using an expanding window.

4.4 XGBoost

XGBoost (Chen and Guestrin, 2016) uses level-wise tree growth. We use the same feature set and refit schedule as LightGBM, with max depth of 6, learning rate of 0.05, subsample ratio of 0.8, and early stopping after 30 rounds of no improvement.

4.5 Ensemble

We construct a simple equal-weight ensemble of three models: LightGBM, HAR-RV, and GARCH(1,1). These three are chosen because they are structurally diverse (a non-parametric ML model, a linear RV model, and a parametric time-series model), and forecast combination theory suggests that diversity of base forecasters drives ensemble performance (Timmermann, 2006). The ensemble forecast is:

$$\hat{\sigma}_t^{\text{Ens}} = \frac{1}{3}\left(\hat{\sigma}_t^{\text{LGB}} + \hat{\sigma}_t^{\text{HAR}} + \hat{\sigma}_t^{\text{GARCH}}\right) \tag{7}$$

We deliberately do not optimize the ensemble weights, as Patton (2019) show that equal-weight combinations are typically competitive with or superior to weight-optimized combinations out of sample, because the weight estimation introduces additional noise.

4.6 Evaluation Protocol

Loss functions. We evaluate models using three criteria:

1. **QLIKE** (Patton, 2011): $\text{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{\sigma_t^2}{\hat{\sigma}_t^2} - \ln\frac{\sigma_t^2}{\hat{\sigma}_t^2} - 1\right)$

   This is the theoretically preferred loss function for volatility forecasting because it is robust to noise in the volatility proxy (Patton, 2011). Lower is better.

2. **RMSE**: $\text{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\sigma_t - \hat{\sigma}_t)^2}$

3. **Mincer-Zarnowitz** R^2: The R^2 from regressing realized volatility on the forecast. A well-calibrated forecast has intercept zero, slope one, and high R^2.

Expanding window. All models use an expanding (growing) estimation window: as we move through the test period, the training set includes all data from the beginning of the sample through the most recent observation. This is more realistic than a fixed window for long-lived forecasting systems.

No look-ahead bias. Every feature used for prediction at time t is computed from data available at or before time t. Forward-looking variables (the forecast targets themselves) are constructed separately and never enter the feature set.

5 Results

5.1 Main Results: 5-Day Forecast Horizon

Table 4 reports the out-of-sample performance of the seven model configurations (six individual models plus the ensemble) for the primary forecast target, 5-day forward realized volatility, over the test period January 2022 through November 2025.

Table 4: Out-of-sample volatility forecasting performance (5-day horizon, 2022–2025, $N = 980$ trading days). QLIKE and RMSE: lower is better. MZ R^2: higher is better. Best value in each column is in **bold**.

Model	**QLIKE**	**RMSE**	**MAE**	**MZ R^2**
GARCH(1,1)	0.3806	0.0796	0.0523	0.2835
EGARCH(1,1)	0.3748	0.0769	0.0520	0.3097
GJR-GARCH	0.3447	**0.0734**	0.0489	**0.3802**
HAR-RV	0.4198	0.0779	0.0507	0.2895
LightGBM	0.3632	0.0742	0.0476	0.3754
XGBoost	0.3553	0.0745	**0.0470**	0.3638
Ensemble	**0.3431**	0.0738	0.0475	0.3515

Several patterns are worth highlighting.

GJR-GARCH is the best individual model on the full test sample. This was not what we expected going in. The asymmetric GARCH specification, which gives negative returns a larger weight in the variance equation, outperforms both tree-based methods and the standard GARCH on QLIKE (0.3447 vs. 0.3806), delivers the lowest RMSE (0.0734), and achieves the highest Mincer-Zarnowitz R^2 (0.3802). The leverage effect, it turns out, is not just a well-known empirical regularity; it is one of the most valuable pieces of information for forecasting volatility over this test period. The 2022 bear market, with its sharp, asymmetric drawdowns, rewarded models that respond aggressively to negative returns. We caution that this full-sample ranking masks substantial regime dependence, which we examine in the subperiod analysis below and in Section 6.

Tree-based methods are strong but do not dominate. LightGBM and XGBoost both outperform standard GARCH and EGARCH, confirming that conditioning on a richer feature set (VIX, lagged RV at multiple horizons, leverage indicators) improves forecasts. XGBoost edges out LightGBM on QLIKE (0.3553 vs. 0.3632) and produces the lowest MAE in the table (0.0470, narrowly ahead of GJR-GARCH's 0.0489), though the differences are small. On QLIKE and RMSE, however, neither tree model beats GJR-GARCH. This suggests that the asymmetric volatility response that GJR-GARCH captures parametrically with a single indicator term is the dominant signal, and the 30-feature panel available to the tree models adds only marginal value on top.

HAR-RV underperforms, and the regime-conditional analysis in Section 6 explains why. The three-parameter linear model, which has been remarkably competitive in prior studies, ranks last on full-sample QLIKE (0.4198). The regime-conditional decomposition in Table 7 shows that the underperformance is concentrated in the high-RV quartile, where HAR-RV's QLIKE balloons to 0.9015 because a linear regression on past RV cannot extrapolate to days of extreme realized volatility. In the lower-RV quartiles HAR-RV is the fifth-best of seven (QLIKE 0.2592), only modestly behind GJR-GARCH (0.2502) and competitive with the front of the field. HAR conditions only on past realized volatility and has no access to VIX or to asymmetric response terms, so when the RV level moves outside the convex hull of the training distribution the linear projection breaks. This is a known weakness of OLS on non-stationary RV; the 5-minute realized-volatility variants of HAR studied in the high-frequency literature partially address it, but require intraday data that we deliberately do not use here.

The ensemble wins on QLIKE. The equal-weight average of LightGBM, HAR-RV, and GARCH(1,1) achieves the lowest QLIKE of any model (0.3431), edging out GJR-GARCH by a narrow margin. This result is consistent with the forecast combination literature (Timmermann, 2006): combining structurally diverse forecasters (a parametric time-series model, a linear realized-volatility model, and a nonparametric ML model) smooths idiosyncratic errors. The ensemble does not have the highest MZ R^2 (GJR-GARCH does), which means the ensemble is better calibrated on average but slightly less correlated with realized volatility point-by-point. For risk management applications where calibration matters more than correlation, the ensemble is the safer choice.

5.2 Feature Importance

Table 5 reports the top ten features by LightGBM split-based importance from a single fit on the full training+validation window (2004–2021), with early stopping on the last 20% of that window. The model converged at 80 boosting rounds.

Table 5: Top 10 features by LightGBM split-importance (single fit on 2004–2021 train+validation window, $n_{\text{boost}} = 80$). Share is split count divided by total split count across all 30 features.

Rank	**Feature**	**Splits**	**Share**
1	VIX close, day t	213	8.9%
2	Daily log return, day t	198	8.3%
3	Negative return, 5-day cumulative	154	6.4%
4	5-day RV (lag 22)	148	6.2%
5	HAR monthly component	147	6.1%
6	VIX 5-day average (lag 1)	145	6.0%
7	Volume / 22-day MA ratio	131	5.5%
8	5-day RV (lag 5)	113	4.7%
9	Positive return, 5-day cumulative	108	4.5%
10	5-day RV (lag 10)	102	4.3%

Three observations on this ranking. First, VIX is the most-split feature. The VIX is the market's forward-looking expectation of 30-day volatility, derived from S&P 500 option prices, and aggregates information about upcoming FOMC meetings, earnings season, and geopolitical events that no backward-looking feature can capture. That LightGBM identifies VIX as its most valuable input is sensible: the model is effectively learning to combine a market-implied forecast with statistical features from past data. The 5-day VIX average (rank 6) provides a smoother version of the same signal.

Second, the leverage effect is captured through three different features: today's log return (rank 2), the cumulative negative return over the past five days (rank 3), and the cumulative positive return over the past five days (rank 9). Together these three features account for roughly 19% of total split count, suggesting the tree model spends a substantial fraction of its capacity learning the same asymmetric response that GJR-GARCH encodes structurally with two parameters.

Third, the long-horizon RV signals dominate among the lagged-RV features (lag 22 at rank 4, the HAR monthly component at rank 5, lag 10 at rank 10), while the daily and weekly HAR components do not crack the top ten. The model treats short-horizon RV as already absorbed by the more recent return-based features, and uses the long-horizon channels to anchor the forecast level. Volume relative to its 22-day moving average (rank 7) earns a place in the top ten, consistent with the practitioner intuition that abnormal volume precedes abnormal volatility, though it is not as dominant as the price-based features.

5.3 Subsample Analysis: A Surprising Reversal

The full-sample ranking masks substantial regime dependence. To examine this, we split the test period two ways: by calendar year (2022 standalone vs. 2023–2025) and by volatility regime (top-quartile RV days vs. the remaining three quartiles). Section 6 reports the full numerical table; the headline pattern is striking enough to deserve discussion here.

In 2022, the worst year of the test period for equity markets, EGARCH is the best individual model with QLIKE of 0.2346, followed by GARCH(1,1) at 0.2579 and GJR-GARCH at 0.2597. The tree-based models perform notably worse: XGBoost at 0.3148 and LightGBM at 0.3429, with LightGBM ranking last among the seven. The ensemble lands in the middle at 0.2616. In the calmer 2023–2025 period, the ranking inverts: tree models lead (XGBoost 0.3693, LightGBM 0.3702), the ensemble is competitive at 0.3712, and GJR-GARCH (0.3740) leads the GARCH family but trails the trees, with HAR-RV last at 0.4568.

This inversion is the central subtle finding of the paper. Tree-based ML models struggle in 2022 not because the 2022 vol level is unprecedented (the training data includes the 2008 financial crisis and the March 2020 COVID crash, both of which delivered higher RV than 2022) but because the immediate 2019–2021 validation period that drives early stopping and final feature importances was unusually calm. The trees' parameters are tuned to feature interactions that worked in the post-COVID-shock recovery, and 2022 is the first sustained departure from that calm regime. Parametric GARCH models, by contrast, have only three or four parameters fit on the full 2004–2021 history, which is dominated by long-run variance dynamics rather than recent regime tuning. In 2023–2025, when the RV level normalizes and intra-year regimes look more like the training distribution, the larger tree feature panel begins to pay off and tree models overtake.

The full-sample QLIKE is therefore a weighted average over two qualitatively different forecasting environments. A practitioner who reads only the full-sample ranking would conclude that GJR-GARCH is the best individual model, when in fact GJR-GARCH is third in 2022 and fourth in 2023–2025. The ensemble's robustness comes from precisely this regime-averaging property: it is rarely best in any one period, but never disastrously wrong either.

6 Robustness

We examine the robustness of our main findings along three dimensions: subperiod stability by calendar year, subperiod stability by realized-volatility regime, and statistical significance of forecast differences.

6.1 Subperiod Analysis: Calendar Years

We first split the test period at the end of 2022. The 2022 subsample (251 trading days) covers the bear market year, with the S&P 500 down 19.4% and 5-day realized volatility frequently exceeding 20% annualized. The 2023–2025 subsample (729 trading days) covers the recovery, the AI-driven rally, and the subsequent rate-normalization regime.

Table 6: QLIKE by calendar-year subperiod (lower is better; **bold** = best in column).

Model	**2022 (high vol, N=251)**	**2023–2025 (lower vol, N=729)**	**Full sample (N=980)**
GARCH(1,1)	0.2579	0.4228	0.3806
EGARCH	**0.2346**	0.4230	0.3748
GJR-GARCH	0.2597	0.3740	0.3447
HAR-RV	0.3123	0.4568	0.4198
LightGBM	0.3429	0.3702	0.3632
XGBoost	0.3148	**0.3693**	0.3553
Ensemble	0.2616	0.3712	**0.3431**

The headline finding from this split is the model-class reversal across the two subperiods. In 2022, the GARCH family wins: EGARCH leads at QLIKE 0.2346, with GARCH(1,1) and GJR-GARCH within four basis points. The ensemble follows at 0.2616. The tree-based models trail substantially: XGBoost at 0.3148, LightGBM at 0.3429 (last). In 2023–2025 the order inverts: tree models lead (XGBoost 0.3693, LightGBM 0.3702, ensemble 0.3712 close behind), GJR-GARCH leads the GARCH family at 0.3740, and HAR-RV is last at 0.4568.

The interpretation we favor is that 2022 was structurally out-of-distribution for models trained on 2004–2021. The high-volatility regime did not look like the training data: a GARCH model with a small parameter footprint adapts smoothly to the new variance level, whereas a 30-feature LightGBM relies on feature interactions that do not generalize. In 2023–2025 the regime is closer to historical norms, the larger feature panel pays off, and the trees overtake. The ensemble does not win either subperiod individually, but is never disastrously wrong, which is what averaging structurally diverse forecasters delivers.

6.2 Subperiod Analysis: Volatility Regime

A second split conditions on the realized volatility level itself rather than the calendar. We classify each test-day as *high* or *lower* regime based on whether actual 5-day RV exceeds the 75th percentile of test-period RV (threshold 19.05% annualized). Compared with the calendar split, this isolates regime effects from secular calendar dynamics.

Table 7: QLIKE by realized-volatility regime, top-quartile RV vs. remaining three quartiles. **Bold** = best in column.

Model	**High vol (top quartile, N=245)**	**Lower vol (bottom 3Q, N=735)**
GARCH(1,1)	0.6913	0.2770
EGARCH	**0.6252**	0.2913
GJR-GARCH	0.6282	0.2502
HAR-RV	0.9015	0.2592
LightGBM	0.7276	0.2418
XGBoost	0.6841	0.2457
Ensemble	0.6850	**0.2291**

Three patterns emerge. First, in the high-RV quartile, the GARCH family clusters at the top: EGARCH leads at QLIKE 0.6252, GJR-GARCH essentially tied at 0.6282, GARCH(1,1) at 0.6913. The asymmetric specifications hold up best on tail-volatility days, but the gap to the symmetric GARCH(1,1) is small. HAR-RV collapses to QLIKE 0.9015 in this quartile because the linear regression on past RV cannot extrapolate to days of extreme realized volatility; this single failure mode dominates HAR-RV's weak full-sample showing. LightGBM is second-worst at 0.7276 in the high-vol quartile, consistent with the calendar-year picture in 2022 where it ranked last.

Second, in the lower-volatility quartiles, the ordering inverts and the ensemble wins clearly: Ensemble 0.2291, LightGBM 0.2418, XGBoost 0.2457, GJR-GARCH 0.2502, HAR-RV 0.2592, GARCH 0.2770, EGARCH 0.2913. The ensemble's roughly 130-basis-point lead over LightGBM is larger than any spread in the high-vol quartile, and the GARCH-family models, especially EGARCH, become the worst performers. This is the mirror image of the high-vol picture: EGARCH was best when RV was extreme and is now last when RV is moderate.

Third, the regime split decomposes HAR-RV's full-sample weakness cleanly. HAR-RV ranks fifth out of seven in the lower-vol quartiles (QLIKE 0.2592), only modestly behind GJR-GARCH at 0.2502, but its high-vol QLIKE of 0.9015 is so much worse than the field that it drags the full-sample average to last place. Knowing this, a practitioner who deploys HAR-RV could either gate it off when VIX exceeds a threshold or combine it with a tail-aware backstop, both of which are out of scope here but follow directly from the regime decomposition.

These two splits tell a consistent story: the full-sample QLIKE ranking averages over qualitatively different forecasting environments, and a practitioner who reads only the full-sample number is missing where each model fails. The ensemble's claim to robustness is sharpened: it ranks fourth in the high-vol quartile but first in the lower-vol quartile, which is a more defensible profile than any single model's "best-on-average" win.

6.3 Statistical Significance

We test whether the QLIKE differences between key model pairs are statistically meaningful using the Diebold and Mariano (1995) test, applied to the loss differential series $d_t = L(\sigma_t, \hat{\sigma}_t^A) - L(\sigma_t, \hat{\sigma}_t^B)$ where L is the QLIKE loss. We use a HAC (Newey–West) standard error with bandwidth equal to the forecast horizon minus one ($h - 1 = 4$). The null hypothesis is equal expected loss; reported p-values are two-sided.

Table 8: Diebold–Mariano two-sided test on full-sample QLIKE loss differentials ($N = 980$). Negative mean diff means model A beats B.

Pair A vs. B	**Mean diff**	**t-stat**	**p-value**	**Conclusion**
GJR-GARCH vs. GARCH(1,1)	−0.0358	−2.14	0.033	GJR-GARCH, sig. 5%
GJR-GARCH vs. EGARCH	−0.0301	−2.00	0.045	GJR-GARCH, sig. 5%
GJR-GARCH vs. HAR-RV	−0.0751	−2.99	0.003	GJR-GARCH, sig. 1%
Ensemble vs. GARCH(1,1)	−0.0374	−3.25	0.001	Ensemble, sig. 1%
Ensemble vs. GJR-GARCH	−0.0016	−0.12	0.903	Indistinguishable
Ensemble vs. LightGBM	−0.0201	−0.96	0.337	Indistinguishable
LightGBM vs. HAR-RV	−0.0566	−1.48	0.140	Indistinguishable
XGBoost vs. LightGBM	−0.0079	−0.92	0.359	Indistinguishable
EGARCH vs. GARCH(1,1)	−0.0058	−0.84	0.399	Indistinguishable

Three takeaways from Table 8. First, the GJR asymmetric specification is significantly better than the other GARCH variants at the 5% level, and significantly better than HAR-RV at the 1% level: the leverage effect is statistically meaningful, not just economically meaningful. Second, the ensemble is statistically indistinguishable from GJR-GARCH ($p = 0.90$), so the headline result that "the ensemble wins on QLIKE" should be read as "the ensemble matches GJR-GARCH while being more robust across subperiods," not as a clear-cut victory. Third, LightGBM does not beat HAR-RV at conventional significance levels ($p = 0.14$), and the tree models do not separate from each other (XGBoost vs. LightGBM, $p = 0.36$). The visual rankings in Table 4 look cleaner than the statistics actually support.

The practical implication: if forced to choose a single model, GJR-GARCH is a defensible choice and is statistically as good as the ensemble on the full sample. If a practitioner cares about robustness across regimes (the subperiod tables above), the ensemble is the safer choice, accepting that the difference is not significant on any one period.

6.4 Joint and Multiple-Model Tests: MCS, SPA, and Reality Check

The pairwise Diebold–Mariano tests in Table 8 are exposed to multiple-testing inflation: running DM on every pair of models in the panel inflates the family-wise error rate above the nominal α. Three significance frameworks are designed to address this by jointly assessing the entire panel: the Hansen-Lunde-Nason Model Confidence Set (Hansen et al., 2011), the Hansen Superior Predictive Ability test (Hansen and Lunde, 2005), and the White Reality Check (White, 2000). We compute all three using the open-source `vol-eval` package of Khan (2026b), with stationary-bootstrap resampling (Politis and Romano, 1994), the QLIKE loss function (Patton, 2011), HAC bandwidth $h - 1 = 4$, and 2000 bootstrap replications.

Model Confidence Set. Table 9 reports the MCS at 90% and 95% confidence on full-sample QLIKE.

Table 9: Hansen-Lunde-Nason Model Confidence Set (MCS) on full-sample 5-day QLIKE losses ($N = 980$). The MCS is the set of models that cannot be rejected as containing the best forecaster at the stated confidence level. Computed with `vol-eval` 0.1.0 (Khan, 2026b), 2000 stationary-bootstrap replications, $t_{\max}$ statistic.

MCS confidence level	Survivors (out of 7 models)
90% MCS	6: GARCH, EGARCH, GJR-GARCH, LightGBM, XGBoost, Ensemble
95% MCS	7: all models survive

The 90% MCS eliminates only HAR-RV (elimination p-value 0.060), consistent with HAR-RV's full-sample QLIKE of 0.4198 being clearly worse than the field. At the 95% confidence level even HAR-RV is not eliminated. The MCS thus tells a sharper version of the same story as the pairwise DM tests: with the exception of HAR-RV at the 90% level, the field cannot reject equal expected loss across the panel. The "ensemble narrowly leads" observation in Section 5 is a point-estimate statement, not a separation in distribution.

Superior Predictive Ability and Reality Check. Table 10 reports SPA (Hansen and Lunde, 2005) and Reality Check (White, 2000) p-values for three candidate "benchmarks": HAR-RV (the standard linear benchmark for RV), GARCH(1,1) (the standard parametric benchmark), and GJR-GARCH (our best individual model). The SPA and Reality Check nulls are that the benchmark is not outperformed by any competitor in the panel; rejecting the null means some alternative is significantly better.

Table 10: Hansen SPA test (Hansen and Lunde, 2005) and White Reality Check (White, 2000) p-values on full-sample 5-day QLIKE losses ($N = 980$, 2000 stationary-bootstrap replications, computed with `vol-eval` 0.1.0 (Khan, 2026b)). Low p-values reject the null that the benchmark is the best.

Benchmark	**SPA** p_c	**SPA** p_l	**SPA** p_u	**Reality Check** p
HAR-RV	0.003	0.003	0.003	0.003
GARCH(1,1)	0.008	0.008	0.008	—
GJR-GARCH	0.529	0.529	0.820	—

The pattern is clean. Both HAR-RV and GARCH(1,1) are statistically dominated by some model in the panel ($p < 0.01$); the Reality Check confirms HAR-RV at the same significance level. GJR-GARCH, in contrast, is not dominated ($p_c = 0.53$, $p_u = 0.82$), confirming what the pairwise DM table had already shown: GJR-GARCH is statistically indistinguishable from the ensemble and is, in this sense, the model the panel cannot improve upon. A practitioner who reads Table 4 as "the ensemble is the best" should read Tables 9 and 10 as "the ensemble matches the best individual model, but the panel as a whole speaks with one voice—once you go beyond HAR-RV and standard GARCH, no model significantly dominates."

Running the MCS, SPA, and Reality Check is now a five-line operation against an open-source package with audit-grade tests (Khan, 2026b). The under-reporting of these joint tests in published vol-forecasting comparisons documented in Khan (2026b) is therefore not a tooling problem; it is an editorial-norms problem.

6.5 Limitations and Extensions

We note four limitations that future work should address. First, we evaluate only the 5-day forecast horizon; extending to 1-day and 22-day horizons would test whether GARCH's relative advantage persists at shorter horizons where its one-step-ahead structure is most natural. Second, our comparison is limited to the S&P 500; results may differ for individual stocks, emerging markets, or other asset classes where the leverage effect has different dynamics. Third, we exclude recurrent neural networks (LSTM, GRU) and transformer architectures from the final comparison; the deferral was driven by toolchain compatibility in our development environment, not by a methodological judgment, and the question of whether properly implemented sequence models add value over the methods we test here remains open. Fourth, the GARCH evaluation refits parameters on an expanding window through the end of each batch (Section 4); a strict walkforward design that holds parameters fixed within each batch is more conservative and would slightly inflate the GARCH QLIKE values, though the qualitative rankings should be unaffected over a 44-day refit cadence.

7 Conclusion

This paper ran a controlled horse race across seven volatility-forecasting configurations (six individual models plus an equal-weight ensemble) on S&P 500 data from 2004 through 2025, with out-of-sample evaluation over the 2022–2025 period. The results admit a three-sentence summary that hides as much as it reveals, so we offer it twice. The simple version: GJR-GARCH is the best individual model on the full test sample, the tree methods (LightGBM, XGBoost) are strong runners-up, and an equal-weight ensemble of LightGBM, HAR-RV, and GARCH(1,1) beats everything on QLIKE by a hair. The honest version: the full-sample QLIKE ranking averages over two qualitatively different forecasting environments, the 2022 GARCH-friendly regime and the 2023–2025 tree-friendly regime, and the ensemble's win over GJR-GARCH is not statistically significant ($p = 0.90$).

A few reflections beyond the numbers.

The strength of GJR-GARCH on the full sample is a reminder that a well-specified parametric model can beat machine learning when the parametric assumption happens to match the data-generating process. The leverage effect is real, persistent, and well-captured by the GJR asymmetric term. ML models can learn this asymmetry from data (the negative-return features in our tree models do something similar), but they have to discover it from noisy examples rather than encoding it structurally. In a test period heavy on asymmetric downside risk, the parametric advantage wins on average.

The underperformance of HAR-RV deserves comment. HAR has been a perennial strong performer in the volatility-forecasting literature, often matching or beating ML models. Its full-sample last-place finish here (QLIKE 0.4198) is driven by collapse in the high-RV quartile (QLIKE 0.9015), where the linear regression on past RV cannot extrapolate to days of extreme volatility. In the lower three RV quartiles HAR-RV is competitive (QLIKE 0.2592, fifth of seven). HAR conditions only on past realized volatility and has no access to VIX or to asymmetric response terms. In calmer markets and at horizons where high-frequency RV is feasible, HAR would likely be more competitive.

The ensemble result is the most practically useful finding, with a caveat. A risk manager who maintains three structurally diverse models and averages their forecasts achieves QLIKE statistically indistinguishable from the best individual model, while being notably more robust to regime change. The ensemble was never first in any subperiod we examined, but never disastrously last either. That is what averaging structurally diverse forecasters delivers, and it is the version of the "ensembles win" result we recommend reporting in practitioner contexts.

A closing word on limitations. The comparison is limited to a single asset (the S&P 500), and the results may not generalise to individual stocks, emerging markets, or other asset classes. We use daily data rather than intraday data; a HAR model fed 5-minute realized volatility would likely outperform our daily version. Transformer architectures are absent from the test, despite some recent promise, because they require substantially more data and compute to train than the rest of the field. The 5-day forecast horizon is the only one we evaluate; the ranking at the 1-day or 22-day horizon is a separate empirical question. Finally, our test period, while deliberately challenging, is still only four years long, so the regime-dependence story above is identified from two macro environments (2022 vs. 2023–2025) rather than many.

Reproducibility

This paper aims to score 2 on the Reproducibility Disclosure Score rubric proposed in Khan (2026a): *(i) code public*, *(ii) data openly accessible*. The full LaTeX source, the data-collection and modeling scripts, the feature panel, the per-day forecasts of all seven model configurations, and the metrics used to populate every table in this paper are released at github.com/ayk5511/volatility-forecasting under permissive licenses (MIT for code, CC0 for the derived datasets). The raw inputs (`^GSPC` and `^VIX` OHLCV) are obtainable for free from Yahoo Finance via the included `01_collect_data.py` script. We invite corrections, extensions, and challenges to the rankings reported here through the repository's issue tracker.

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